

Topic to discuss

- Gauss Forward difference Interpolation
- Gauss Forward Table
- Gauss Forward Formula
- Numerical Problem
- Homework Problem

Gauss Forward Difference Interpolation

A central difference Interpolation method for equally spaced data.

It express $y(x)$ as a series in $u = \frac{x - x_0}{h}$ using forward central differences from a difference table centered near x , giving good accuracy for interior points close to x_0 .

When to use :

- When x lies just greater than the chosen central entry. (for x just above x_0)
- Use it when x lies between x_0 and x_1
i.e, $0 < u < 1$

Gauss Forward Table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-2}	y_{-2}				
x_{-1}	y_{-1}	$\Delta y_{-2} = y_{-1} - y_{-2}$	$\Delta^2 y_{-2} = \Delta y_{-1} - \Delta y_{-2}$	$\Delta^3 y_{-2} = \Delta^2 y_{-1} - \Delta^2 y_{-2}$	
x_0	y_0	$\Delta y_{-1} = y_0 - y_{-1}$	$\Delta^2 y_{-1} = \Delta y_0 - \Delta y_{-1}$	$\Delta^3 y_{-1} = \Delta^2 y_0 - \Delta^2 y_{-1}$	$\Delta^4 y_{-2} = \Delta^3 y_{-1} - \Delta^3 y_{-2}$
x_1	y_1	$\Delta y_0 = y_1 - y_0$	$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$		
x_2	y_2	$\Delta y_1 = y_2 - y_1$			

Gauss Forward difference Interpolation Formula

The Formula is :

$$y(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{(u+1)u(u-1)}{3!} \Delta^3 y_0 + \frac{(u+1)u(u-1)(u-2)}{4!} \Delta^4 y_0 \dots$$

where, $u = \frac{x - x_0}{h}$

and h is constant difference
between successive values of x .

Also, $0 < u < 1$

Numerical Problem

Q: from the following table, find the value of $e^{1.17}$ using Gauss Forward Formula.

x	e^x
1.00	2.7183
1.05	2.8577
1.10	3.0042
1.15	3.1582
1.20	3.3201
1.25	3.4903
1.30	3.6693

Solution: step length : $h = 1.05 - 1.00$
 $= 0.05$

Choose central value closest to $x(1.17)$; $x_0 = 1.15$

Interpolation point : $x = 1.17$

$$\therefore u = \frac{x - x_0}{h} = \frac{1.17 - 1.15}{0.05}$$

$$u = 0.4$$

Gauss Central difference Table ($x_0 = 1.15$)

	x	$y = f(x)$ e^x	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$\Delta^6 y$
x_{-3}	1.00	2.7183	0.1394	0.0071	0.0004			
x_{-2}	1.05	2.8577	0.1465	0.0075	0.0004	0		
x_{-1}	1.10	3.0042	0.154	0.0079	0.0004	0	0	
x_0	1.15	3.1582	0.1619	0.0083	0.0005	0.0001	0.0001	0.0001
x_1	1.20	3.3201	0.1702	0.0088				
x_2	1.25	3.4903	0.179					
x_3	1.30	3.6693						

Gauss Central difference Interpolation Formula,

$$y(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{3!} \Delta^3 y_{-1} + \frac{(u+1)u(u-1)(u-2)}{4!} \Delta^4 y_{-2} +$$

$$\frac{(u+2)(u+1)u(u-1)(u-2)}{5!} \Delta^5 y_{-2} + \frac{(u+2)(u+1)u(u-1)(u-2)(u-3)}{6!} \Delta^6 y_{-3}$$

$$= 3.1582 + 0.4 \times 0.1619 + \frac{0.4(0.4-1)}{2!} \times 0.0079 + \frac{(0.4+1)0.4(0.4-1)}{3!} \times 0.0004 +$$

$$+ \frac{(0.4+1)0.4(0.4-1)(0.4-2)}{4!} \times 0 + \frac{(0.4+2)(0.4+1)0.4(0.4-1)(0.4-2)}{5!} \times 0.0001$$

$$+ \frac{(0.4+2)(0.4+1)0.4(0.4-1)(0.4-2)(0.4-3)}{6!} \times 0.0001$$

$$= 3.1582 + 0.06476 - 0.000948 - 0.0000224 + 0.000001075 - 0.000000466$$

$$= 3.221990209$$

Ans.

Homework Problem

Q: Find $\sin 42^\circ$ using Gauss central difference from the following table.

$x (^\circ)$	35	40	45	50	55
$\sin x$	0.573576	0.642788	0.707107	0.766044	0.819152

Solution : Step length, $h = 40 - 35 = 5$

Central point (closest to 42°): $x_0 = 45$

Interpolation point, $x = 42$

$$\therefore u = \frac{x - x_0}{h} = \frac{42 - 45}{5} = -0.6$$

Gauss Forward Central Difference Table,

	x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-2}	35	0.573576	0.069212	-0.004893	-0.000489	
x_{-1}	40	0.642788	0.064319	-0.005382	-0.000447	0.000042
x_0	45	0.707107	0.058937	-0.005829		
x_1	50	0.766044	0.053108			
x_2	55	0.819152				

Gauss Forward Central Difference formula,

$$y(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{3!} \Delta^3 y_{-1} + \frac{(u+1)u(u-1)(u-2)}{4!} \Delta^4 y_{-2}$$

$$= 0.707107 + (-0.6) \times 0.058937 + \frac{-0.6(-0.6-1)}{2!} \times -0.005382 +$$

$$\frac{(-0.6+1)(-0.6)(-0.6-1)}{3!} \times -0.000447 + \frac{(-0.6+1)(-0.6)(-0.6-1)(-0.6-2)}{4!} \times 0.000042$$

$$= 0.707107 - 0.0353622 - 0.00258336 - 0.000028608 - 0.000001747$$

$$= 0.669131085$$

Ans

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